

Galaxy Training: A powerful framework for teaching!

Helena Rasche¹, Saskia Hiltemann², Björn Grüning³, Galaxy Training Network, Bérénice Batut^{4,5}

¹Department of Clinical Genetics, Leiden University Medical Center (LUMC), 2333 ZA Leiden, The Netherlands, ²Central Data Facility, University of Freiburg, 79104 Freiburg, Germany, ³Department of Bioinformatics, University of Freiburg, Freiburg, Germany, ⁴CNRS, Institut Français de Bioinformatique, IFB-Core, UAR 3601, ⁵Plateforme AuBi, Mésocentre Clermont-Auvergne, Université Clermont Auvergne.

Corresponding authors: berenice.batut@france-bioinformatique.fr.

Training

There is an ongoing explosion of scientific datasets being generated, brought on by recent technological advances in many areas of the natural sciences. As a result, the life sciences have become increasingly computational in nature, and bioinformatics has taken on a central role in research studies. However, **basic computational skills, data analysis, and stewardship are still rarely taught** in life science educational programs, resulting in a skills gap in many of the researchers tasked with analyzing these big datasets.

TIaaS

Training Infrastructure as a Service (TIaaS) is a **free** service providing a **dedicated job queue** for the participants of training events, in order to reduce waiting times on the cluster and ensure courses run smoothly and efficiently.

Since 2018:

860+
Events

38k+
Students

The TIaaS service also provides instructors with a **dashboard**, enabling them to monitor the progress of the participants.

A collection of FAIR training materials

The Galaxy Training Network (GTN) has developed the Galaxy Training Platform (<https://training.galaxyproject.org>), an **open access, community-driven** framework for the collection of **FAIR training materials** for data analysis utilizing the user-friendly Galaxy framework as its primary data analysis platform.

35+
Topics

470+
FAQs

565k+
Visits/year

450+ Tutorials

Rich metadata

Structured content

Theory and analysis explanation
✂ Hands-on to apply the concepts
📄 Formative assessment question

Strong technical support

🌐 Galaxy servers with needed tools
📄 Available data
🔄 Workflow
📺 Recordings
🔍 Frequently Asked Questions

Community oriented

🗣 Translation
👤 Highlighted contributors and funding
📄 Embedded feedback form
💬 Community support chat

490+
Contributors

3970+
Feedback

& many other features:

Learning paths, Notebooks, Video slides, Automatic assessability, Choose Your Own Adventures, etc

While initially aimed at supporting researchers directly, the GTN framework has proven to be an **invaluable resource for educators** as well. We have focused our efforts in recent years on adding increased **support** for this growing community of instructors.

Massive, online & worldwide training events with Galaxy Training Academy

Since 2021, GTN organizes Galaxy Training Academy: a **free, online, self-paced**, and **choose-your-own-adventure** style training events.

220+
Videos

140+
Hours
of recordings

130+
Instructors

10k+
Registrations

This year we have partnered with BioConductor and the RO-crates team to deliver bioinformatics training that goes beyond the Galaxy!

Acknowledgments

universität freiburg

Erasmus MC
University Medical Center Rotterdam

Erasmus

Gallantries



With the support of the
Erasmus+ Programme
of the European Union



Galaxy
PROJECT



The Institut Français de Bioinformatique (IFB) is funded by France 2030, via the Agence Nationale de la Recherche grants ANR-11-INBS-0013(PIA1 RENABI-IFB), and ANR-21-ESRE-0048 (PIA3 ESR/EquipeX+ MUDIS4LS).



Icons by miftakhudin, icon 54, didik, V12n, Mustofa Bayu, Ricons, Skena Rafis, Riyan Resdian, MD Studio, Icon Z, bajuja, feri ulan taufig, Iconiqu, Subaida, Thomas Deckert from Noun Project